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Simulating Personalized Treatment Pathways in Chronic Disease Management Using Reinforcement Learning and Synthetic Patient Data

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CLINICAL & AGENTIC AI TAKEAWAY

Reinforcement learning and synthetic patient data can improve personalized treatment plans for chronic disease management, reducing unsafe behavior and enhancing patient outcomes.

ABSTRACT

We have created a method for developing personalized treatment plans using synthetic patient data and reinforcement learning; these methods can improve physician clinical decisions based on a patient's symptoms, adherence to their prescribed treatments and safety concerns. Our results show that as our model learns, it selects progressively better treatment options than before, avoids unsafe behavior, and helps patients across all characteristics (or patient profiles). Ablation studies also confirm that tracking symptoms and the prescribed medication schedule were key elements in achieving positive patient outcomes compared to other methodologies that are based on rules. Our trained model outperformed other models based on rules regarding reward, safety and symptom improvement. While the data used in the simulation environment, the results provide a level of realism that should aid clinicians wishing to develop new and safer treatment options.

Study Overview & Background

We have created a method for developing personalized treatment plans using synthetic patient data and reinforcement learning; these methods can improve physician clinical decisions based on a patient's symptoms, adherence to their prescribed treatments and safety concerns. Our results show that as our model learns, it selects progressively better treatment options than before, avoids unsafe behavior, and helps patients across all characteristics (or patient profiles). Ablation studies also confirm that tracking symptoms and the prescribed medication schedule were key elements in achieving positive patient outcomes compared to other methodologies that are based on rules. Our trained model outperformed other models based on rules regarding reward, safety and symptom improvement. While the data used in the simulation environment, the results provide a level of realism that should aid clinicians wishing to develop new and safer treatment options.

Methodology & Clinical Focus

Investigation into artificial intelligence models, machine learning algorithms, and clinical predictive tools applied to orthopedic surgery and musculoskeletal medicine.

Key Findings & Analysis

Reinforcement learning and synthetic patient data can improve personalized treatment plans for chronic disease management, reducing unsafe behavior and enhancing patient outcomes.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** Deepak Upadhyay, Nookala Venu, Deepak Suresh Asudani, Ansh Vadhera, Priyanshi Semwal, K. Sharma
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